

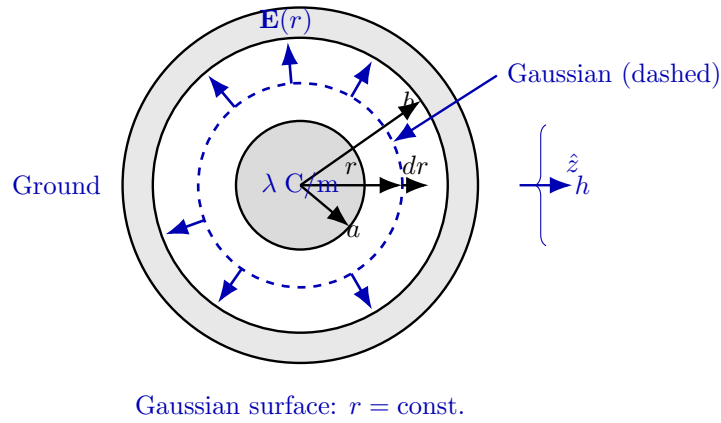
1 Gauss's law (electrostatics).

$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{encl}}}{\epsilon_0} \iff \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (1)$$

Example 1: Two concentric conducting cylinders (coaxial geometry)

Problem set-up. Two long, concentric *conducting* cylinders:

- Inner conductor radius a carries a uniform *line charge density* λ (C/m).
- Outer conductor has inner radius b ($b > a$). The region $a < r < b$ is air/vacuum.
- We find $\mathbf{E}(r)$ in the region $a < r < b$ (and note the behavior elsewhere).



Solution (Gauss surface). By cylindrical symmetry, $\mathbf{E}$ is purely radial and depends only on r :

$$\mathbf{E}(r) = E(r) \hat{\mathbf{r}}.$$

Choose a coaxial Gaussian cylinder of radius r and length h in the region $a < r < b$.

- On the **curved surface**, $\hat{\mathbf{n}} = \hat{\mathbf{r}}$ and $E(r)$ is constant:

$$\iint_{\text{side}} \mathbf{E} \cdot d\mathbf{S} = \iint_{\text{side}} E(r) \hat{\mathbf{r}} \cdot (\hat{\mathbf{r}} dS) = E(r) (2\pi r h).$$

- On the **top and bottom caps**, $d\mathbf{S} = \pm \hat{\mathbf{z}} dS$ while $\mathbf{E} \parallel \hat{\mathbf{r}}$, so $\mathbf{E} \cdot d\mathbf{S} = 0$:

$$\iint_{\text{caps}} \mathbf{E} \cdot d\mathbf{S} = 0.$$

Therefore, the total flux through the closed Gaussian surface is

$$\oiint_S \mathbf{E} \cdot d\mathbf{S} = E(r) (2\pi r h).$$

The enclosed charge is the charge on the inner conductor over length h :

$$Q_{\text{encl}} = \lambda h.$$

Gauss's law gives

$$E(r) (2\pi r h) = \frac{\lambda h}{\epsilon_0} \implies \boxed{\mathbf{E}(r) = \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad a < r < b.}$$

Electric field inside a conductor: $\mathbf{E} = 0$ for $r < a$ (electrostatics)

If $\mathbf{E} \neq 0$ inside a conductor, free electrons must drift (Drude picture)

A conductor contains mobile (nearly free) electrons. If an electric field exists inside the conducting material, each electron experiences

$$\mathbf{F} = -e \mathbf{E}.$$

A simple microscopic model (Drude) includes collisions with an average collision time τ (or collision frequency $\nu = 1/\tau$). The equation of motion is modeled as

$$m_e \frac{d\mathbf{v}}{dt} = -e \mathbf{E} - m_e \frac{\mathbf{v}}{\tau}. \quad (2)$$

Taking *time-average* (steady state), $\langle d\mathbf{v}/dt \rangle = 0$, gives

$$\langle \mathbf{v} \rangle \equiv \mathbf{v}_d = -\frac{e\tau}{m_e} \mathbf{E}. \quad (3)$$

Thus a nonzero $\mathbf{E}$ in a conductor produces a nonzero drift velocity $\mathbf{v}_d$.

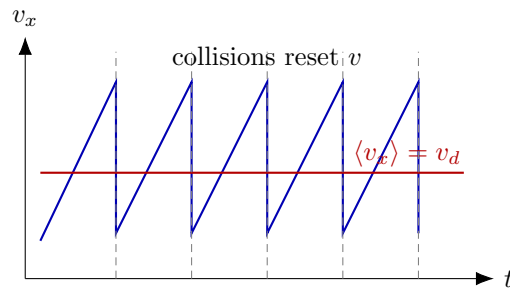


Figure 1: Cartoon of electron velocity component under a constant $\mathbf{E}$ with frequent collisions: acceleration between collisions, randomization at collisions, but a nonzero average drift v_d .

Current density and why electrostatics forces $\mathbf{E} = 0$ inside the conductor

If the electron number density is n_0 (electrons per m^3), then current density is

$$\mathbf{J} = -n_0 e \mathbf{v}_d. \quad (4)$$

Using (3):

$$\mathbf{J} = \left(\frac{n_0 e^2 \tau}{m_e} \right) \mathbf{E} \equiv \sigma \mathbf{E}. \quad (5)$$

So if $\mathbf{E} \neq 0$ inside a conductor, then $\mathbf{J} \neq 0$ (a DC current must flow).

Electrostatics means *charges are not continually moving in steady DC flow inside the bulk* (no sustained current in the interior unless an external source maintains it). Therefore, in electrostatic equilibrium,

$$\mathbf{J} = 0 \implies \mathbf{E} = 0 \text{ inside the conducting material.}$$

Equivalently: the conductor must be an equipotential in electrostatics (potential constant in the bulk).

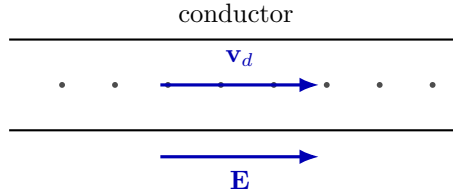


Figure 2: If a static $\mathbf{E}$ existed in the conductor, electrons would drift, producing $\mathbf{J} = \sigma \mathbf{E}$. In electrostatic equilibrium (no sustained current in the bulk), $\mathbf{J} = 0 \implies \mathbf{E} = 0$ inside.

Alternate way of explaining the same: Why $J_r = 0$ inside the inner conductor ($r < a$), hence $E_r = 0$. Consider the *metal* region of the inner conductor, $r < a$. In electrostatics (steady state), charges do not continuously flow through the bulk; otherwise charge density inside the metal would keep changing with time.

Step 1: A radial current would imply charge accumulation. Assume, for contradiction, that there is a nonzero radial current density inside the metal:

$$J_r(r) \neq 0 \text{ for some } r < a.$$

Pick a thin cylindrical Gaussian “pillbox” volume completely inside the metal between radii r and $r + \Delta r$ and height h (any h). The only way current can leave/enter this volume is through its *curved radial surfaces* (the end caps are normal to $\hat{\mathbf{z}}$).

If $J_r \neq 0$, there is a net current crossing the curved surfaces. By charge conservation (continuity equation),

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0. \quad (6)$$

A nonzero net outward/inward current through the boundary forces

$$\nabla \cdot \mathbf{J} \neq 0 \Rightarrow \frac{\partial \rho}{\partial t} \neq 0,$$

meaning charge density inside the metal would change with time (charge would pile up somewhere).

But electrostatics means steady state: $\partial \rho / \partial t = 0$ everywhere in the conductor bulk. Therefore we must have

$$\nabla \cdot \mathbf{J} = 0 \quad \text{and specifically no sustained radial flow in the bulk: } J_r = 0 \text{ for } r < a.$$

Step 2: Ohm's law in a conductor gives $E_r = 0$. Inside a conducting material (locally),

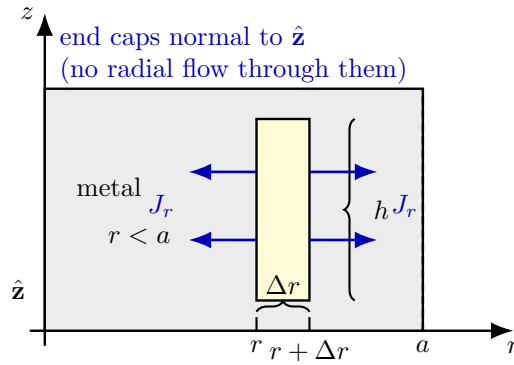
$$\mathbf{J} = \sigma \mathbf{E}.$$

Taking the radial component,

$$J_r = \sigma E_r.$$

Since electrostatic equilibrium requires $J_r = 0$ in the bulk,

$$E_r = 0 \quad \text{for } r < a.$$



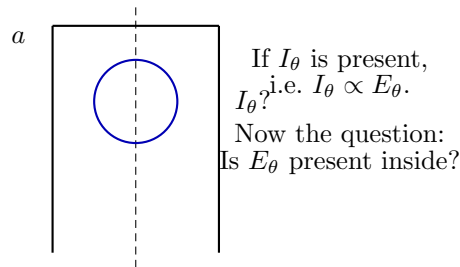
If $J_r \neq 0$ in the bulk, net current crosses the curved surface

$$\Rightarrow \nabla \cdot \mathbf{J} \neq 0 \Rightarrow \frac{\partial \rho}{\partial t} \neq 0 \text{ (charge accumulation).}$$

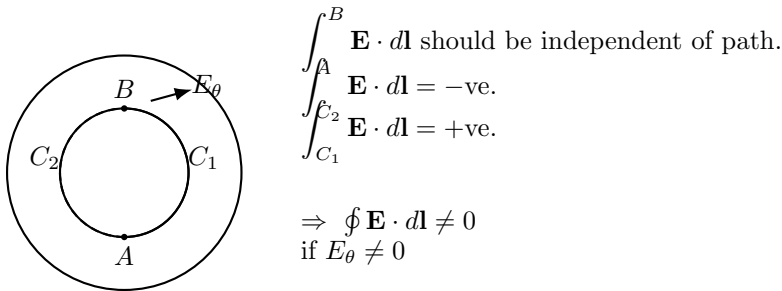
$$\text{Electrostatics: } \frac{\partial \rho}{\partial t} = 0 \Rightarrow J_r = 0 \text{ in the bulk.}$$

$$\text{Ohm: } J_r = \sigma E_r \Rightarrow E_r = 0 \text{ for } r < a.$$

Now, can current flow in θ direction without requiring replacement of e^- ? i.e. is it possible to have circulating current in electrostatic condition here?



Take cross-section;



$\Rightarrow$ Cannot have circulating current in electrostatics.

$\Rightarrow \oint \mathbf{E} \cdot d\mathbf{l} = 0 \Rightarrow E_\theta = 0 \Rightarrow I_\theta = 0.$

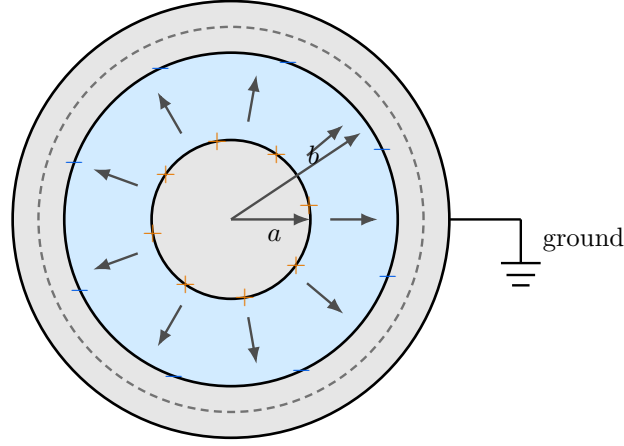
Precautionary note: This is the case for electrostatics.

Later we will introduce Faraday's law, where a time-varying magnetic field can produce a nonzero $\oint \mathbf{E} \cdot d\mathbf{l}$ and hence circulating currents.

Hence, in summary:

Electric field is zero inside a conducting material (electrostatics).

Why we must discuss the region $r > b$ carefully (and what actually happens)



$\mathbf{E} \neq 0$ only for $a < r < b$; $\mathbf{E} = 0$ in metals and for $r > b$ (grounded case).

Figure 3: Grounded coaxial geometry: induced charge (–) appears on the inner surface of the outer conductor, confining the field to the gap.

Because the outer cylinder is a *conductor connected to ground*, it is not an isolated object: it can exchange charge with the earth. In electrostatics a conductor satisfies two key facts:

1. The electric field inside the conducting material is zero: $\mathbf{E} = 0$ in the metal.
2. Any net charge given to a conductor resides on its surface(s), arranged so that the conductor is an equipotential.

Step 1: What charge must appear on the inner surface of the grounded outer cylinder?

Consider a Gaussian cylinder of length L lying *within the metal* of the outer conductor (i.e., choose r such that you are inside the conducting material). Since $\mathbf{E} = 0$ everywhere on that Gaussian surface, Gauss's law gives

$$\oint \mathbf{E} \cdot d\mathbf{S} = 0 \implies Q_{\text{enclosed}} = 0.$$

But this Gaussian surface encloses:

- the charge on the inner conductor, which is λL ,
- plus the charge induced on the *inner surface* of the outer conductor, call it $Q_{\text{inner surf}}$.

Hence,

$$\lambda L + Q_{\text{inner surf}} = 0 \implies Q_{\text{inner surf}} = -\lambda L.$$

So the grounded outer cylinder must induce *exactly* $-\lambda$ (C/m) on its inner surface.

Step 2: What about charge on the *outer* surface, and what does that imply for $r > b$?

Let the charge on the *outer* surface of the outer cylinder be $Q_{\text{outer surf}}$ (per length L). The *net* charge on the outer conductor is then

$$Q_{\text{cond}} = Q_{\text{inner surf}} + Q_{\text{outer surf}}.$$

For an *isolated* (ungrounded) outer conductor, Q_{cond} would be fixed (often zero), forcing $Q_{\text{outer surf}} = +\lambda L$, which would create a nonzero field for $r > b$.

However, the outer conductor is *grounded*, meaning it is electrically connected to the earth, a very large charge reservoir. As a result, the conductor can *take in or give away charge* until its potential becomes $V = 0$. Therefore, it is not forced to keep any particular net charge on itself. In an ideal coaxial geometry, the induced charge $-\lambda$ appears on the inner surface to cancel the field inside the metal, and any additional charge that would otherwise appear on the outer surface simply flows to (or from) ground. The physically relevant outcome is:

$$Q_{\text{outer surf}} = 0 \quad \Rightarrow \quad Q_{\text{enclosed (for } r > b)} = \lambda L + (-\lambda L) + 0 = 0.$$

Then Gauss's law immediately yields

$$\mathbf{E}(r) = \mathbf{0} \quad \text{for } r > b.$$

Field in all regions (typical coax, outer conductor cancels external field).

$$\mathbf{E}(r) = \begin{cases} \mathbf{0}, & r < a \quad (\text{inside inner conductor}), \\ \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, & a < r < b \quad (\text{air gap}), \\ \mathbf{0}, & r > b \quad (\text{inside/ outside outer conductor if it carries induced } -\lambda \\ & \text{on its inner surface and no net external charge}). \end{cases}$$

Interpretation (important)

- The field in the *dielectric region* $a < r < b$ is nonzero and is the usual coaxial field:

$$\mathbf{E}(r) = \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, \quad a < r < b,$$

because a Gaussian cylinder of radius r (with $a < r < b$) encloses only λL .

- The grounded outer conductor induces $-\lambda$ on its *inner* surface, terminating the field lines inside the coaxial gap.
- Because grounding allows charge exchange with the earth, the outer surface need not carry $+\lambda$; it can carry *zero* net charge, so the external field vanishes.

For a long, ideal coaxial structure with the outer conductor grounded, the net enclosed charge for any Gaussian surface with $r > b$ is zero. Hence

$$\mathbf{E}(r) = \mathbf{0} \quad \text{for } r > b.$$

Potential in a coaxial cable from the electric field

From the previous result for an ideal coax (inner conductor radius a , outer conductor inner radius b),

$$\mathbf{E}(r) = \begin{cases} \mathbf{0}, & r < a \quad (\text{inside inner conductor}), \\ \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}}, & a < r < b \quad (\text{dielectric/air gap}), \\ \mathbf{0}, & r > b \quad (\text{outside, for grounded outer conductor}). \end{cases}$$

Step: integrate $\mathbf{E}$ to get potential. Choose the potential at the inner conductor surface as a reference:

$$\phi(a) = V_0.$$

For any r in the gap ($a < r < b$),

$$\phi(r) - \phi(a) = - \int_a^r \mathbf{E} \cdot d\mathbf{l} = - \int_a^r \frac{\lambda}{2\pi\epsilon_0 r'} dr'.$$

Hence,

$$\boxed{\phi(r) = V_0 - \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r}{a}\right), \quad a < r < b.}$$

Grounded outer conductor (most common case). If the outer conductor is grounded, then $\phi(b) = 0$. Substituting $r = b$ gives

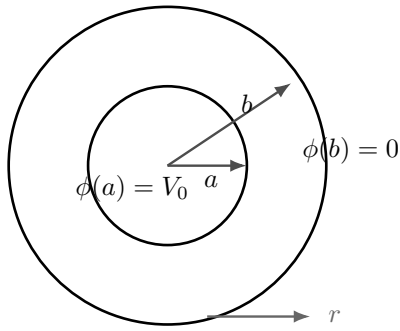
$$0 = V_0 - \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right) \Rightarrow \boxed{V_0 = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right).}$$

Therefore the potential in the gap can also be written as

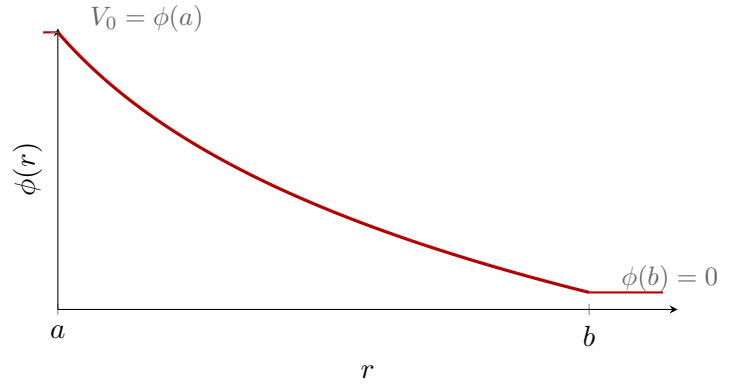
$$\boxed{\phi(r) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{r}\right), \quad a < r < b.}$$

Potential everywhere (grounded outer conductor).

$$\phi(r) = \begin{cases} V_0, & r \leq a, \\ \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{r}\right), & a < r < b, \\ 0, & r \geq b. \end{cases}$$



(a) Coax: reference potentials at $r = a$ and $r = b$.



(b) Potential decreases logarithmically in the gap and is constant in conductors.

Figure 4: From the coaxial electric field $E_r \propto 1/r$ in the gap, the potential varies as $\phi(r) = V_0 - \frac{\lambda}{2\pi\epsilon_0} \ln(r/a)$ (and with grounding, $\phi(r) = \frac{\lambda}{2\pi\epsilon_0} \ln(b/r)$).

Application of Gauss's law; Example-2: Field outside a uniformly charged solid sphere

Problem. A solid sphere of radius R carries total charge Q uniformly distributed throughout its volume. Find the electric field for $r > R$.

Key idea. By spherical symmetry, for a Gaussian sphere of radius r (with $r > R$), the electric field has only a radial component and has the same magnitude everywhere on the Gaussian surface:

$$\mathbf{E}(r) = E(r) \hat{\mathbf{r}}, \quad \text{constant on } S_r.$$

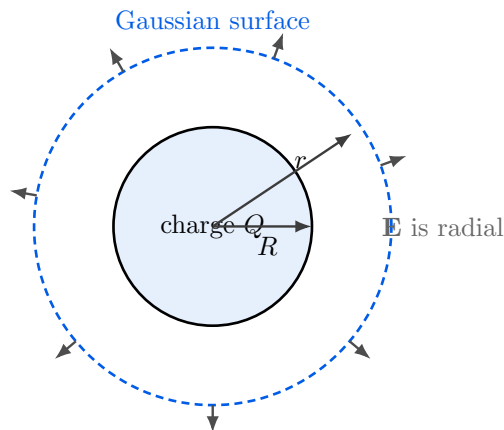


Figure 5: Uniformly charged solid sphere (radius R) and a spherical Gaussian surface of radius $r > R$.

Apply Gauss's law.

$$\oint_{S_r} \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\epsilon_0}.$$

For $r > R$, the Gaussian surface encloses the entire charge, so $Q_{\text{enc}} = Q$. Also $\mathbf{E} \parallel d\mathbf{S}$ and $E(r)$ is constant on S_r , hence

$$E(r) \oint_{S_r} dS = E(r) (4\pi r^2) = \frac{Q}{\epsilon_0}.$$

Therefore,

$$E(r) = \frac{Q}{4\pi\epsilon_0 r^2}, \quad r > R$$

and in vector form,

$$\mathbf{E}(r) = \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}, \quad r > R.$$

Conclusion. *The field outside a uniformly charged solid sphere is the same as if all the charge were concentrated at the center.*

Divergence of the field of a point charge at the origin (Dirac delta as a distribution)

Field of a point charge. A point charge Q at the origin has electric field

$$\mathbf{E}(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0} \frac{\mathbf{r}}{r^3}, \quad r = \|\mathbf{r}\|.$$

For $r \neq 0$ this field is smooth and one can compute directly that

$$\nabla \cdot \mathbf{E}(\mathbf{r}) = 0 \quad (r \neq 0).$$

However, the field is *singular* at the origin, so $\nabla \cdot \mathbf{E}$ cannot be treated as an ordinary function everywhere. The correct statement uses the **Dirac delta** as a **distribution**.

Dirac delta in 3D: distribution definition

The **3D Dirac delta** $\delta^{(3)}(\mathbf{r})$ is defined by its action on any smooth test function $\varphi(\mathbf{r})$ of compact support:

$$\int_{\mathbb{R}^3} \delta^{(3)}(\mathbf{r}) \varphi(\mathbf{r}) d^3r = \varphi(\mathbf{0}).$$

It is *not* an ordinary function; it is a **distribution**.

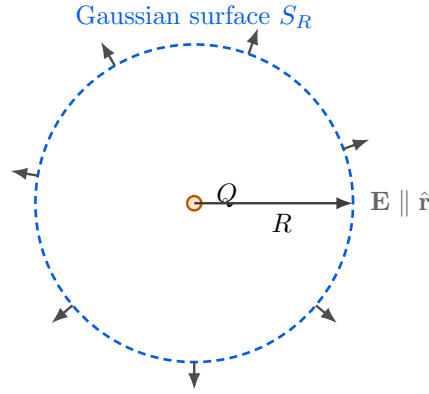


Figure 6: Point charge at the origin and a spherical Gaussian surface (drawn as a cross-section). Flux is nonzero even though $\nabla \cdot \mathbf{E} = 0$ for $r \neq 0$, so the divergence must include a delta term at the origin.

Key identity (distributional divergence)

(Proof: Not included here):

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) = 4\pi \delta^{(3)}(\mathbf{r}).$$

Then multiplying by $\frac{Q}{4\pi\epsilon_0}$ immediately gives

$$\nabla \cdot \mathbf{E} = \frac{Q}{\epsilon_0} \delta^{(3)}(\mathbf{r}).$$

Equivalently, the charge density of a point charge is

$$\rho(\mathbf{r}) = Q \delta^{(3)}(\mathbf{r}) \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}.$$

Integrating over any volume V containing the origin,

$$\int_V \nabla \cdot \mathbf{E} d^3r = \frac{1}{\epsilon_0} \int_V \rho d^3r \Rightarrow \oint_{\partial V} \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0},$$

which recovers Gauss's law (integral form) for a point charge.